ductors, holes are the *majority carriers* and thermally generated electrons are the *minority carriers.* In this case, p is quite large as compared to n but the law of mass-action still holds, i.e., $np = n_i p_i = n_i^2$. It may also be noted that in either type of semiconductors, the overall charge neutrality is maintained as no charge is added to or removed from the material.

7.4 DRIFT VELOCITY, MOBILITY AND CONDUCTIVITY OF INTRINSIC SEMICONDUCTORS

At finite temperatures, due to thermal agitation and lattice vibrations, some of the valence band electrons are always present in the conduction band, i.e., at ordinary temperatures, an intrinsic semiconductor always contains some free electrons in the conduction band and an equal number of holes in the valence band. In the absence of any applied electric field, these electrons and holes move in random directions and constitute no current. When an electric field is applied, these electrons and holes get accelerated towards the opposite ends of the field and their velocity begins to increase. This increase in velocity, however, does not continue indefinitely because of the collisions of these carriers with the various types of obstacles, such as atomic nuclei, phonons, etc., present in the semiconductor. Depending on the mean free path, the carriers acquire an average increment in velocity which is lost during the subsequent collision. This extra velocity acquired by the carriers in the presence of an applied electric field is called the *drift velocity* and is denoted by v_d. It is proportional to the strength $\mathcal{E}$ of the applied electric field, i.e.,

$$v_d \propto \mathcal{E}$$

or

$$v_d = \mu E \tag{7.3}$$

where the constant μ is called the *mobility* of the charge carrier and is defined as the drift velocity acquired by a carrier per unit electric field strength.

In an intrinsic semiconductor, since the electrons move in nearly empty conduction band while holes move in nearly full valence band, the properties such as mobility, conductivity, etc. of electrons are, in general, different from those of holes. Let v_{dn}, μ_n and n denote the drift velocity, mobility and concentration of electrons respectively in the conduction band. Then current density due to electrons is given by

$$J_n = nev_{dn} \tag{7.4}$$

where e is the electronic charge. We can write (7.3) for the electrons as

$$v_{dn} = \mu_n \mathcal{E}$$

Therefore, from Eq. (7.4), we obtain

$$J_n = ne\mu_n \mathcal{E} \tag{7.5}$$

Comparing it with Ohm's law, i.e.,

$$J_n = \sigma_n \mathcal{E},$$

where σ_n represents the electronic conductivity of the material, we get

$$\sigma_n = ne\mu_n \tag{7.6}$$

Similarly, we can write the expression for the conductivity due to holes in the valence band as

$$\sigma_p = pe\mu_p \tag{7.7}$$

where p and μ_p represent the concentration and mobility of holes respectively. Thus the total conductivity of the material is

$$\begin{aligned} \sigma &= \sigma_n + \sigma_p \\ &= e\,(n\mu_n + p\mu_p) \end{aligned} \tag{7.8}$$

For an intrinsic semiconductor,

$$n = p = n_i$$

Therefore, Eq. (7.8) becomes

$$\sigma = en_i\,(\mu_n + \mu_p) \tag{7.9}$$

It is important to note that, in semiconductors, the movement of carriers or the flow of current is, in fact, the consequence of the following two processes :

(i) drift of carriers under the effect of an applied field; the resulting current is called the *drift current.*

(ii) diffusion of carriers under the effect of concentration gradient of dopants present inside the semiconductor ; the corresponding current is called the *diffusion current.*

In the above treatment, we have considered only the drift current contribution. The diffusion current contribution is absent in semiconductors having a uniform distribution of impurities.

Variation of Conductivity with Temperature

Assuming mobilities to be independent of temperature, the temperature dependence of conductivity arises because of the variation of intrinsic carrier concentration, n_i, with temperature. It will be proved that n_i is given by

$$n_i = \frac{2(2\pi kT)^{3/2}}{h^3} \left(m_n^* m_p^*\right)^{3/4} \exp\left(-\frac{E_g}{2kT}\right) \tag{7.10}$$